Robert Poznański

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Education

Technikum Elektryczno-Elektroniczne nr 7 im. Powstańców Śląskich Sep 2023 — Present
Technik Programista *Kraków, Poland*

- Average grade: 5.59
- **INF.03** qualification exam – *passed* (100%, 94%)[1]
- **Erasmus+** internship in Game Development and 3D Modeling (Unreal Engine, Blender), Granada, Spain
- Relevant coursework: Algorithms, OOP, Databases, Web Development, Computer Networks

Experience

Software Developer Intern Jul 2026
Cavatina *Kraków, Poland*

- Built a backend for an internal AI chat powered by Google Gemini, with centralized API key management
- Built a React frontend for managing employee access, monitoring token usage, and interacting with the AI chat
- Automated invoice processing using Tesseract OCR to extract data from PDF documents

Game Development Intern May 2026
Arcan Studios *Granada, Spain*

- Learned 3D modeling, rigging, and character animation using Blender
- Imported custom models and animations into an Unreal Engine game
- Programmed character movement, level obstacles, and gameplay logic using Unreal Engine Blueprints

Projects

Custom 16-bit RISC Cpu Emulator github.com/din0x/custom-risc16-emulator

- Designed a custom Instruction Set Architecture with variable-length encoding
- Written an assembler for the custom instruction set
- Written the emulator for the RISC16 instruction set in the C Programming Language

Stratospheric Balloon

- Implemented SPI/USART drivers for u-blox GPS modules, LoRa SX1278 receivers/transmitters
- Built reliable telemetry communication over LoRa for long-range data transmission
- Designed packaging protecting electronics at 30 km, −50 °C, 13 hPa

UBX Protocol (de)serializer Library github.com/din0x/ubx

- Developed a Rust library implementing the UBX protocol used by GNSS receivers
- Provides packet encoding and stream decoding with automatic synchronization and recovery
- Designed for robust communication in embedded and navigation applications

AVR HAL Library github.com/din0x/avr

- Developed a Hardware Abstraction Layer for ATmega microcontrollers
- Provides safe and ergonomic access to AVR peripherals
- Focused on type safety and embedded systems reliability

WebGPU Renderer github.com/din0x/plotrs

- Built a reusable 2D and 3D rendering engine on top of wgpu/WebGPU
- Implemented graphics abstractions and rendering utilities in Rust
- Used as a foundation for graphics and visualization projects

Skills

- **Languages**: English (B2+)[2], Polish (Native)
- **Programming Languages**: Rust, Python, Haskell, C, JavaScript, Bash, SQL,
- **Tools**: Git, KiCad
- **Embedded Systems**: AVR Microcontrollers, LoRa, u-blox, Embedded Rust, Hardware Abstraction Layers

[1] Scored 100% on the theoretical exam and 94% on the practical exam.
[2] United Kids language school diploma; level is an approximate CEFR equivalent.